

Volatility 3 — Plugin Reference

Memory Forensics Plugin Guide, organized by analysis category

1. Process Analysis

Examines running/hidden processes and their relationships.

Plugin	Description
windows.pslist	Lists active processes in memory, including PID, PPID, process name, and process-related timestamps.
windows.pstree	Displays processes in a parent-child hierarchy to show which process created another process.
windows.psscan	Scans memory for process structures and can identify processes that may no longer appear in the normal process list.
windows.psxview	Compares multiple process-enumeration methods to identify inconsistencies that may indicate hidden processes.
windows.cmdline	Displays command-line arguments associated with processes.
windows.cmdscan	Attempts to recover command history from Windows command shells.
windows.consoles	Provides information about Windows console sessions and command activity.

2. Network Analysis

Reveals network connections and listening sockets in memory.

Plugin	Description
windows.netscan	Scans memory for network connections, listening sockets, local/remote IP addresses, ports, protocols, states, and associated processes.
windows.netstat	Displays network connection information recovered from Windows networking structures.

3. DLL / Module Analysis

Lists loaded DLLs and kernel modules, incl. hidden ones.

Plugin	Description
windows.dlllist	Lists DLLs loaded by a specific process or processes.
windows.ldrmodules	Examines different Windows loader lists to identify inconsistencies in DLL/module visibility.
windows.modscan	Scans memory for kernel module structures, including modules that may not appear in normal enumeration.
windows.modules	Lists loaded Windows kernel modules and drivers.
windows.driverscan	Scans memory for Windows driver objects.
windows.drivermodule	Cross-references driver objects against loaded modules to detect rootkit-hidden drivers.

4. Malware / Injection Analysis

Detects code injection, hooks, and malicious memory regions.

Plugin	Description
windows.malfind	Searches process memory for potentially suspicious memory regions commonly associated with code injection or malicious executable memory.
windows.vadyarascan	Applies YARA rules to process virtual memory regions identified through VAD structures.
windows.yarascan	Searches memory for strings or patterns using YARA rules.
windows.mutantscan	Searches memory for mutant objects, which can provide useful information during malware investigations.
windows.bigpools	Examines large kernel memory pool allocations that may contain useful forensic artifacts.
windows.poolscanner	Scans memory pools for specific Windows kernel object types.

5. Virtual Memory Analysis

Examines a process's virtual address space (VAD tree).

Plugin	Description
windows.vadinfo	Displays Virtual Address Descriptor information describing the virtual memory regions of a process.
windows.vadwalk	Walks the VAD tree of a process to examine its virtual memory allocations.
windows.memmap	Displays the memory mappings associated with a process.
windows.vadinfo	Helps identify memory regions, their protection properties, and mapped objects within a process.
windows.vadwalk	Provides a lower-level view of the virtual address space and its VAD hierarchy.

6. File System / File Analysis

Finds and recovers file objects referenced in memory.

Plugin	Description
windows.filescan	Scans memory for Windows file objects and can reveal files referenced by the system or processes.
windows.dumpfiles	Attempts to extract cached file contents from memory.
windows.mftscan	Searches memory for Master File Table-related structures and artifacts.
windows.fileobj	Provides information about Windows file objects associated with processes.

7. Registry Analysis

Inspects registry hives, keys, and values in memory.

Plugin	Description
windows.registry.hivelist	Lists registry hives present in memory and their virtual/physical locations.
windows.registry.hivescan	Scans memory for registry hive structures.
windows.registry.printkey	Displays registry keys and values from a specified registry hive.
windows.registry.userassist	Extracts UserAssist registry artifacts that can provide evidence of applications executed by users.
windows.registry.certificates	Extracts certificate-related information from Windows registry data.

8. Persistence Analysis

Uncovers services, tasks, and registry-based persistence.

Plugin	Description
windows.svcscan	Scans memory for Windows service objects and provides service configuration information.
windows.scheduled_tasks	Examines scheduled-task information that may reveal persistence or automated execution.
windows.registry.userassist	Provides evidence of programs executed through user activity artifacts.
windows.registry.printkey	Allows investigation of registry locations commonly associated with persistence mechanisms.

9. Process Handles / Objects

Lists handles processes hold to system objects.

Plugin	Description
windows.handles	Lists handles opened by processes to files, registry keys, processes, threads, events, mutexes, and other Windows objects.
windows.getsids	Displays security identifiers associated with processes and their tokens.
windows.privileges	Displays privileges associated with process tokens.
windows.mutantscan	Searches for mutant objects, which can sometimes reveal malware synchronization mechanisms.
windows.symlinkobj	Examines symbolic-link objects present in the Windows object manager.

10. User / Session Analysis

Shows user sessions, SIDs, and logon context.

Plugin	Description
windows.sessions	Lists Windows sessions and provides information about users and session contexts.
windows.getsids	Shows SIDs associated with processes, helping identify their security context.
windows.envvars	Displays environment variables associated with processes.
windows.logonsessions	Examines Windows logon-session information.
windows.cachedump	Attempts to recover cached domain authentication artifacts from memory.

11. Security / Credential Analysis

Extracts password hashes and cached credential secrets.

Plugin	Description
windows.hashdump	Attempts to extract Windows account password-hash information from memory.
windows.cachedump	Attempts to recover cached domain logon information.
windows.lsadump	Attempts to extract LSA-related secret information from memory.
windows.getsids	Identifies security identifiers associated with processes.
windows.privileges	Displays privileges held by process tokens.

12. Kernel / Driver Analysis

Examines kernel modules, drivers, and system tables.

Plugin	Description
windows.modules	Lists kernel modules currently loaded in memory.
windows.modscan	Scans memory for kernel module structures.
windows.driverscan	Searches memory for Windows driver objects.
windows.drivermodule	Cross-references driver objects against loaded modules to detect rootkit-hidden drivers.
windows.driverirp	Examines I/O Request Packet information associated with drivers.
windows.ssdt	Displays the System Service Descriptor Table and associated system-call information.
windows.callbacks	Displays registered Windows kernel callback routines.

13. Process Memory / Injection Support

Deeper process-memory tools supporting injection analysis.

Plugin	Description
windows.malfind	Identifies potentially suspicious executable memory regions in processes.
windows.vadinfo	Displays process virtual memory regions and their protection information.
windows.vadwalk	Walks the process VAD tree to examine memory allocations.
windows.memmap	Displays process memory mappings.
windows.dlllist	Lists DLLs loaded into a process for module investigation.
windows.lldrmodules	Helps identify inconsistencies in module-loading structures.
windows.vadyarascan	Searches process memory regions with YARA rules.

14. System Information

Basic OS, kernel, and memory-image metadata.

Plugin	Description
windows.info	Provides operating-system, kernel, architecture, and memory-image information.
windows.statistics	Provides statistics and information about the memory image and its structures.
windows.crashinfo	Extracts information from Windows crash-dump memory images.

15. Services

Details Windows service configuration and status.

Plugin	Description
windows.svcscan	Scans memory for Windows services and displays service-related information.
windows.getservicesids	Retrieves security identifiers associated with Windows services.
windows.svcdiff	Compares service information between relevant Windows structures where supported.

16. Environment Analysis

Reviews process environment variables and sessions.

Plugin	Description
windows.envvars	Displays environment variables associated with processes.
windows.cmdline	Shows the command-line arguments used to start processes.
windows.sessions	Provides information about Windows sessions and their context.

17. PE / Executable Analysis

Extracts PE headers and metadata from executables/DLLs.

Plugin	Description
windows.pedump	Dumps PE-related information from a process or memory region.
windows.verinfo	Extracts version and metadata information from PE files/modules.
windows.dlllist	Lists PE DLL modules loaded by processes.
windows.dumpfiles	Can recover cached executable or DLL content from memory.

18. Timeline / Activity Investigation

Reconstructs a timeline of system and user activity.

Plugin	Description
windows.pslist	Provides process creation and exit-related information useful for activity reconstruction.
windows.pstree	Shows the sequence and relationship of process creation.
windows.cmdline	Provides evidence of commands executed by processes.
windows.netscan	Provides network activity associated with processes.
windows.userassist	Provides evidence of user application execution.
windows.svcscan	Provides service information useful for persistence investigations.

Core Volatility 3 Categories for an Analyst

Category → main plugins, quick-reference summary

Category	Main Plugins
System Information	windows.info, windows.statistics
Process Analysis	windows.pslist, windows.pstree, windows.psscanner, windows.psxview
Command Analysis	windows.cmdline, windows.cmdscanner, windows.consoles
Network Analysis	windows.netscan, windows.netstat
DLL Analysis	windows.dlllist, windows.lldrmodules
Malware Analysis	windows.malfind, windows.vadyarascanner, windows.yarascanner
Memory Analysis	windows.vadinfo, windows.vadwalk, windows.memmap
File System	windows.filescan, windows.dumpfiles, windows.mftscan
Registry	windows.registry.hivelist, windows.registry.hivescan, windows.registry.printkey, windows.registry.userassist
Persistence	windows.svcscan, windows.scheduled_tasks, registry plugins
Handles / Objects	windows.handles, windows.mutantscan
User / Sessions	windows.sessions, windows.getsids, windows.envvars
Credentials	windows.hashdump, windows.cachedump, windows.lsadump
Kernel / Drivers	windows.modules, windows.modscan, windows.driverscan, windows.driverirp, windows.ssdt
PE Analysis	windows.pedump, windows.verinfo, windows.dlllist